

MICHIKA HIRATSUKA

Vancouver, BC | 604-782-1123 | hiratsukamichika@gmail.com | [LinkedIn:](#) | [Github:](#)

OBJECTIVE:

UBC Bachelor of Computer Science student with a strong interest in computer vision, machine learning, and research-driven problem solving. Experienced in Python-based data analysis and applied machine learning, with coursework in computer vision and software engineering. Interested in developing reliable and efficient vision-based systems and applying machine learning techniques to real-world problems such as structural and environmental applications.

EDUCATION:

- **Bachelor of Computer Science (Second Degree)**

The University of British Columbia, Vancouver, BC | Sep 2024 – Current

Relevant Coursework (UBC)

- CPSC 330: Applied Machine Learning
- CPSC 221: Basic Algorithms and Data Structures
- CPSC 310: Introduction to Software Engineering (Currently taking)
- CPSC 425: Computer Vision (Currently taking)
- ENVR 300: Introduction to Research in Environmental Science (Currently taking)

- **Web & Mobile Application Design & Development (Post-Degree Diploma)**

Langara College, Vancouver, BC | Dec 2018

- **Bachelor of Arts in Education (Developmental Psychology, Elementary School Teaching License in Japan)**

Yokohama National University – College of Education and Human Sciences, Japan | March 2012

RELEVANT TECHNICAL SKILLS:

Languages: Python, JavaScript, TypeScript, HTML/CSS, COBOL

Machine Learning / Data: scikit-learn, pandas, NumPy, SHAP, feature engineering

Computer Vision: SIFT, RANSAC, optical flow, segmentation, basic deep learning

Web / Software: React, Next.js, PostgreSQL

Tools: Git, Docker, VS Code

WORK EXPERIENCE

Software Developer (Remote, Freelance), Mazin, JP | Aug 2020 – Jan 2024

- Debugged and resolved complex issues in factory software systems, significantly improving reliability and performance
- Designed and implemented modular frontend architecture based on atomic design principles
- Collaborated in software design discussions and contributed to improving code quality and testing processes
- Wrote clear technical documentation to support maintainability and team development

...Continued...

Frontend Developer (Remote, Freelance), Grammarly, Vancouver, BC, May 2021 – Aug 2021

- Developed dynamic web interfaces using JavaScript with a focus on performance and maintainability
- Improved data structure and metadata handling to enhance system visibility and accessibility

Frontend Developer (Partially Remote), PNI Digital Media Ltd, Vancouver, BC | June 2019 – June 2021

- Created a content management system easily updated by the marketing team – raised order value by 16% from \$42 to \$49
- Implemented web accessibility in production sites to meet Web Content Accessibility Guidelines (WCAG)
- Proposed UX improvements to designers and new feature ideas to project owners to enhance user experience

JavaScript Developer (Freelance), Softtree Technical Systems, North Vancouver | Jan 2019 – Mar 2019

- Resolved system bugs and improved data handling in internal portals
- Updated a request form in the company website utilizing ASP.NET

Software Developer, NCR Japan LTD, JP | April 2013 – April 2014

- Led development and coding of software system with COBOL, which resulted in a successful upgrade of the system
- Created Oracle Database installation manuals for 20 colleagues; developed/tested 300 dummy data for self-tax-payer application

PROJECTS

Computer Vision Project (CPSC 425 - UBC Coursework)

- Implemented feature matching and homography estimation using SIFT and RANSAC to generate panorama images
- Analyzed motion using optical flow on image sequences
- Applied parameter tuning to improve accuracy and robustness

Full-Stack Software Application (CPSC 310 – UBC Coursework)

- Designed and implemented a RESTful API using TypeScript and Express.js
- Built a data ingestion pipeline to process and validate large datasets (ZIP parsing, JSON processing)
- Implemented query functionality with filtering, sorting, and aggregation logic
- Developed comprehensive unit and integration tests using Mocha and Chai
- Improved system robustness through validation, error handling, and modular architecture
- Collaborated with a teammate to design, implement, and test system features

Environmental Modeling & Research Project (ENVR 300 – UBC Coursework)

- Collaborated in a team to build and analyze a predator–prey model under different environmental scenarios
- Contributed to the development of a research proposal, including problem formulation and methodological design